Aarav Shandilya

High school student seeking an internship or research opportunity in Physics, Computer Science, Engineering, or Quantitative Finance. Driven by a passion for innovation, problem-solving, and applying advanced mathematics, statistics, and programming to real-world challenges. Particularly interested in quantitative modeling, data-driven decision making, and computational approaches to finance. Eager to contribute to meaningful projects, learn from experts, and gain hands-on experience in cutting-edge STEM and quantitative fields.

aaravshandilya@gmail.com
(623) 696-1561
Phoenix, Arizona
LinkedIn
Github

EDUCATION

Paradise Valley High School

August 2024 - Current

CREST Computer Science & CREST Engineering

Class of 2028

Weighted GPA: 5.0

Unweighted GPA: 4.0

PSAT: 1460

SAT: 1520

Current Percentile: 99.97%

Relevant Coursework:

AP Physics 1 (4)

AP Physics C: Mechanics (5)

Honors Computer Science (A)

Honors Introduction to Engineering Design (A)

EXPERIENCE:

Oxford University - Research Lead & Intern
April 2026 - Current

- Investigated the application of quantum optimization methods to address NP-hard computational challenges in bioinformatics, specifically targeting pangenome sequence assembly. Formulated and implemented novel higher-order optimization models to represent complex pangenome graph structures efficiently, reducing variable footprints to fit quantum hardware constraints. Bridged theoretical quantum algorithms with practical genomic workflows to improve the scalability of graph traversal and sequence alignment.

Arizona State University & Dell - Research Intern
June 2025 - Current

- Conducted research on quantum annealing and optimization techniques to tackle NP-hard problems, exploring both theoretical and practical approaches. Developed and tested CPLEX models, benchmarking classical and quantum-inspired methods for solution quality and computational efficiency. Collaborated with Dell, applying advanced problem-solving and analytical skills to generate insights with potential real-world enterprise applications.

Grand Canyon University - Research Intern
August 2025 - Current

- Collaborated on integrating multiple robots with independent and coordinated AI systems, developing projects that explored advanced AI interactions. Assisted in designing curriculum for the Bachelor's program in Artificial Intelligence at GCU, contributing educational insights and practical project frameworks. Supported the development of a pre-therapist application by de-identifying PII and applied AI techniques to a prototype antivirus system capable of detecting malware, strengthening both data security and technical problem-solving skills.

SkillsUSA Region 2 Officer - Parliamentarian
July 2025 - Current

- Served as Parliamentarian, representing 7,000 members across Region 2, ensuring meetings and organizational procedures adhered to proper parliamentary protocol. Developed leadership, communication, and organizational skills while guiding officers and members in effective decision-making and governance. Played a key role in maintaining structured, fair, and efficient operations, strengthening both member engagement and the overall impact of SkillsUSA programs.

Next Frontiers Education Founder and President
July 2025 - Current

- Founded and led a nonprofit organization focused on expanding educational access, overseeing all aspects of launch and operations. Developed the organization's website, designed programs, managed documentation and compliance, and led social media outreach and tutor recruitment. Recruited and directed a team of four high school students, successfully guiding the organization from concept to launch while building leadership, project management, and entrepreneurial skills.

Congressional App Challenge - Ambassador
August 2025 - December 2025

- Represented the Congressional App Challenge as a National Ambassador, promoting student participation through presentations and outreach at schools. Developed strong public speaking, advocacy, and leadership skills while educating students on the value of innovation, technology, and civic engagement through app development.

Honors Principles of Sustainability (A)

Honors Civil Engineering & Architecture

Honors Principles of Engineering (A)

AP Computer Science Principles (4)

AP Psychology (4)

AP Precalculus (5)

AP Calculus AB & BC (5 & 5)

AP Computer Science A (5)

AP Human Geography

AP US History

AP European History (4)

AP Seminar (4)

AP Research

AP Statistics

AP Chemistry

AP English Language & Composition

Paradise Valley Community College

August 2024 - Current

(U/W) GPA: 4.0

Relevant Coursework:

Elementary Spanish I & II (SPA 101 & 102)

Calculus III

Vector Calculus

SKILLS

Python

Java

Artificial Intelligence

C, C++

Vector Databases

JavaScript

HTML/CSS

CPLEX

Gurobi

NJCCIC Cybersecurity Intern

July 2025 - October 2025

- Supported cybersecurity operations through hands-on analysis, research, and applied security practices. Gained experience in threat awareness, incident response concepts, and secure handling of sensitive information while strengthening technical and analytical skills in a real-world cybersecurity environment.

PVHS Cybersecurity Club Founder & President

August 2024 - Current

- Lead a chapter of over 30 students and a group of 4 officers
- Developed communication skills, fundraised hundreds of dollars, and media presence while also fostering a passion in cybersecurity
- Collaborated with Industry Professionals to expand our reach, focus, and chapter capabilities

PVHS Science Olympiad Founder & Secretary

August 2024 - Current

- Lead a chapter of over 30 students and a group of 4 officers
- Advertised, fundraised, and collaborated with other officers in order to create a skilled and functional team before developing our competitive aspect through various competitive events

PVHS SkillsUSA Engineering Chapter Vice President

October 2024 - Current

- Led a chapter of over 400 Students and a group of 3 officers
- Served as Vice President, organizing chapter events, fundraisers, and national conference travel while supporting chapter operations and member engagement. Developed leadership, public speaking, and organizational skills through coordinating large-scale activities and representing the chapter in presentations and outreach efforts.

PROJECTS

Optimizing Malware Family Detection by Leveraging DL-Driven CNNs for AI-Assisted Visual Binary Classification

December 2025 - Current

- Lead Researcher | Collaborated with Grand Canyon University to engineer a CNN-based pipeline converting binary executables into images for malware detection; integrated Grad-CAM for model transparency. Awarded Best at Fair and Best in Category.

Fibrin Clot Architecture in Microgravity: Will Spaceflight Weaken How We Stop Bleeding?

August 2025 - Current

- Co-Principal Investigator for a Student Spaceflight Experiments Program project designing a microgravity study on fibrin clot formation, developing the full experimental protocol, materials plan, and analysis framework for an ISS Type 3 FME investigation. Top 3 in the nation, national finalist. Led a Team of 5 People.

Relativistic Black Hole Structures: Reconstructing Event Horizons and Accretion Disks Using Hubble Data

June 2025 - August 2025

- Recreated black hole structures using NASA Hubble Telescope and Harvard research data, building 2D/3D simulations of accretion disks and event horizons. Applied Python and data analysis techniques to validate models, showcasing skills in scientific computing, ML, and research-driven astrophysical modeling. Individual Project.

Project ECHO - ECG Image to Signal Extraction and Rhythm

Qiskit

Intermediate Spanish

Hindi

Leadership

Communication

CAD Software

Cloud Computing

AWARDS & ACHIEVEMENTS

SSEP - 2nd Place Nationally
Student Spaceflight Experimentation program
December 2025

Project Q2 Research Challenge - 2nd Place internationally
Project Q2 Research & Microsoft
August 2026

Top 25 (Internationally)
Math League
March 2025

Top 10 (Internationally)
International Math Bowl
October 2024

AMC 10/12
Multiple Time AIME Qualifier
2024, 2025

State Champion - Arizona
Purple Comet
April 2025

AP Scholar with Distinction
College Board
July 2025

Engineering Design Competition - Gold, Silver
SkillsUSA
February 2025, February 2026

Astronomy - Multiple Invitationals Winner

Classification System

October 2025 - November 2025

- Developed a Java-based pipeline to extract 2D electrocardiogram (EKG/ECG) waveforms from medical images and perform basic rhythm analysis using signal processing and heuristic peak detection. Implemented image preprocessing, QRS detection, and rhythm classification algorithms to interpret waveform features and support diagnostic insights from static ECG scans. Demonstrated practical application of computational signal analysis for biomedical time-series data and automated interpretation of cardiac indicators. Worked with another Highschooler.

Improving the Adoption of Quantum Vector Annealing in Supply Chain Management: Solving the Travelling Salesman Problem (In Collaboration with Dell)

July 2025 - Current

- Developed CPLEX and quantum annealing models to optimize Traveling Salesman Problem instances, creating and curating datasets for benchmarking classical versus quantum-inspired solutions. Led implementation, analysis, and evaluation of optimization strategies, translating results into actionable insights for supply chain management. Co-authored the majority of the paper, collaborating with academic and industry partners to document methodology, results, and practical implications. Worked with Several Professors and Researchers from Dell.

Current Problems Faced by Animals Held in Captivity in Institutions, and Ways Could Technology be Able to Address Them

November 2025 - February 2026

- Analyzed the different types of problems and issues faced by animals held in captivity at institutions such as Zoos, and the negative effect of such issues on mental health, and what technological & psychological solutions could be used to address them. Worked with a team of 3 other high schoolers.

CERTIFICATIONS

Quantum Enigmas
International Business Machine (IBM)
May 2025

- Foundational understanding of quantum computing, including principles such as quantum superposition, entanglement, and measurement. The individual has considered challenges using the IBM Quantum Composer to map problems onto quantum circuits and explored quantum logic gates.

Information Technology Fundamentals
International Business Machine (IBM)
May 2025

- Demonstrates knowledge of information technology (IT) basics, methodologies of troubleshooting, and the tools and resources that IT professionals use. The individual has a conceptual understanding of computer basics, networking, hardware, software, computer security, and demonstrates experience supporting a customer with a simulated remote connection tool.

Cybersecurity Fundamentals
International Business Machine (IBM)
May 2025

- Demonstrates a foundational understanding of cybersecurity concepts, objectives, and practices. This includes cyber threat groups, types of attacks, social engineering, case studies, overall security strategies, cryptography, and common approaches that organizations take to prevent, detect, and respond to cyber attacks. This also includes an awareness of the job market.

Artificial Intelligence Fundamentals
International Business Machine (IBM)
May 2025

Science Olympiad

November 2025

USACO - Bronze

U.S.A Computing Olympiad

January 2025

Top 50 (Internationally)

MIT Informatics Tournament

March 2025

Top 20 (Internationally)

CALICO Informatics Competition

March 2025

1st Place - Biomedical Engineering

PV Science and Engineering Fair

February 2026

Engineering Design Competition - Bronze

SkillsUSA State

April 2026

- Demonstrates knowledge of artificial intelligence (AI) concepts, such as natural language processing, computer vision, machine learning, deep learning, chatbots, and neural networks; AI ethics; and the applications of AI. The individual has a conceptual understanding of how to run an AI model using IBM Watson Studio.

IT Specialist - Python

Pearson VUE

November 2024

- Demonstrates the ability to design, write, test, and debug Python programs using variables, data types, control structures, functions, and modules. Includes experience with basic automation, file handling, and problem-solving using Python libraries.

IT Specialist - Java

Pearson VUE

December 2025

- Validates proficiency in object-oriented programming with Java, including classes, objects, inheritance, and exception handling. Covers skills in building, compiling, and debugging Java applications while applying logical and structured programming techniques.

IT Specialist - Artificial Intelligence

Pearson VUE

December 2025

- Demonstrates understanding of core AI concepts such as machine learning, neural networks, data modeling, and algorithmic decision-making. Includes knowledge of training models, evaluating results, and recognizing ethical and real-world applications of AI systems.

IT Specialist - JavaScript

Pearson VUE

May 2025

- Confirms foundational knowledge of JavaScript for web development, including variables, functions, loops, events, and DOM manipulation. Demonstrates the ability to create interactive and dynamic web content using client-side scripting.

Autodesk Fusion Certified User

Autodesk

January 2024

- Confirms proficiency in using Autodesk Fusion 360 for 3D design and modeling. Skills demonstrated include 2D sketching, part and assembly modeling, advanced modeling techniques, sculpting, creating 3D threads, editing, and preparing designs for 3D printing, showcasing a strong foundation in both design and practical manufacturing workflows.

VOLUNTEERING (500 hours)

Phoenix Public Library | Youth Committee | Arizona Science Center

- Contributed to organizing educational programs and coordinating community events at the branch level in partnership with the Arizona Science Center. Developed teamwork, communication, and event-planning skills while supporting STEM-focused youth engagement and public outreach initiatives.

REFERENCES

Gil Speyer Professor, ASU – speyer@asu.edu

John Fowler Professor, ASU – john.fowler@asu.edu

Corinne Araza Professor, GCU– Corinne.Araza@gcu.edu

Bhawna Verma Teacher, CS, PVHS– bverma@pvschools.net

Reeni Samuel Teacher, Engineering, PVHS– rsamuel@pvschools.net

Sergii Strelchuk Professor, Oxford University- Sergii.strelchuk@cs.ox.ac.uk

LABORATORY SKILLS

Experimental Design & Protocol Development

- Able to design and implement experiments for physics, biomedical, and spaceflight studies, including microgravity experiments. Experienced in defining hypotheses, variables, and experimental procedures. Still developing expertise in highly specialized or industrial-scale laboratory protocols.

Data Collection, Analysis & Visualization

- Proficient in collecting, cleaning, and analyzing experimental data using Python, MATLAB, and Excel. Able to create meaningful visualizations such as graphs, charts, and 2D/3D simulations to interpret results. Less experienced with large-scale datasets beyond classroom or research project levels.

Quantum Computing & Optimization

- Experienced with Qiskit, CPLEX, and quantum-inspired optimization methods, including modeling NP-hard problems like the Traveling Salesman Problem. Able to benchmark classical vs. quantum approaches and interpret results. Limited hands-on experience with cutting-edge quantum hardware outside academic or simulation environments.

Computational Simulation & Modeling

- Proficient in building physics-based and astrophysical simulations, including spacecraft trajectory optimization and black hole accretion disk reconstruction. Able to implement numerical methods, visualize simulations in 2D/3D, and validate results. Still gaining experience with extremely large-scale simulations or commercial software packages.

Signal Processing & Analysis

- Able to process and analyze biomedical signals, such as extracting ECG waveforms from images and performing rhythm classification using Java and Python. Skilled in implementing preprocessing, peak detection, and heuristic analysis. Limited experience with real-time signal acquisition from physical sensors or clinical-grade equipment.

CAD & 3D Modeling

- Proficient in Autodesk Fusion 360, including part and assembly modeling, sculpting, creating 3D threads, and preparing designs for 3D printing. Able to convert experimental ideas into physical prototypes efficiently. Less experienced with advanced industrial manufacturing workflows or specialized machining techniques.

Laboratory Safety & Documentation

- Experienced in following safety protocols and documenting experiments thoroughly for research projects. Able to maintain clean, organized lab environments and report findings clearly. Less experienced in handling hazardous materials or conducting experiments with strict regulatory oversight.

Collaboration with Academic & Industry Researchers

- Skilled at working in teams with professors, industry partners, and other students to execute research projects effectively. Able to coordinate contributions, communicate findings, and integrate diverse expertise. Limited experience in fully independent, high-budget industrial lab settings.